METHODOLOGY RECAP · MODULES M5 + M6

Coverage estimates

What the modules do · How to read the output

WHAT THEY DO

Turn raw service counts into **coverage** — the share of the people who *needed* a service who actually received it.

TWO PARTS

M5 works out the **target population** (the hard part)

M6 turns it into coverage and fills the years between surveys

KEEP IN MIND

These are **estimates from routine data** — good for trends and comparison, not official national figures.

What "coverage" means

Coverage answers one simple question: **of all the people who needed a service, what share actually got it?**

Take first antenatal visits (ANC1). If **10,000** women were pregnant in a district and **8,000** of them had an ANC1 visit, then coverage is 8,000 ÷ 10,000 = **80%.**

So coverage is always a fraction:

> **coverage = the number served ÷ the number who needed it**

$$\text{Service coverage} = \frac{\text{Population who received the service}}{\text{Population who need the service (target population)}}$$

Numerator comes from DHIS2 data — service volumes after quality correction

Several options for estimating the denominator

The catch

The **top** number is easy — the 8,000 ANC1 visits come straight from HMIS; it's just the service count.

The **bottom** number is the hard part: **nobody counts how many pregnant women (or infants) there are.** No facility files a report saying "10,000 women were pregnant here this year."

So the whole job of these modules comes down to one question — **where do we get that bottom number, the target population?** That is what M5 sets out to estimate.

M5 · ESTIMATING THE POPULATION

How M5 finds the bottom number

Step 1 — borrow it from a survey. Every few years a household survey measures coverage *directly*, by interviewing families. It can tell us that, say, **80% of pregnant women got ANC1**. Now combine that with the count we already have: if the **8,000** ANC1 visits are 80% of all pregnant women, then the total is 8,000 ÷ 0.80 = **10,000 pregnant women**. We've recovered the bottom number we couldn't count.

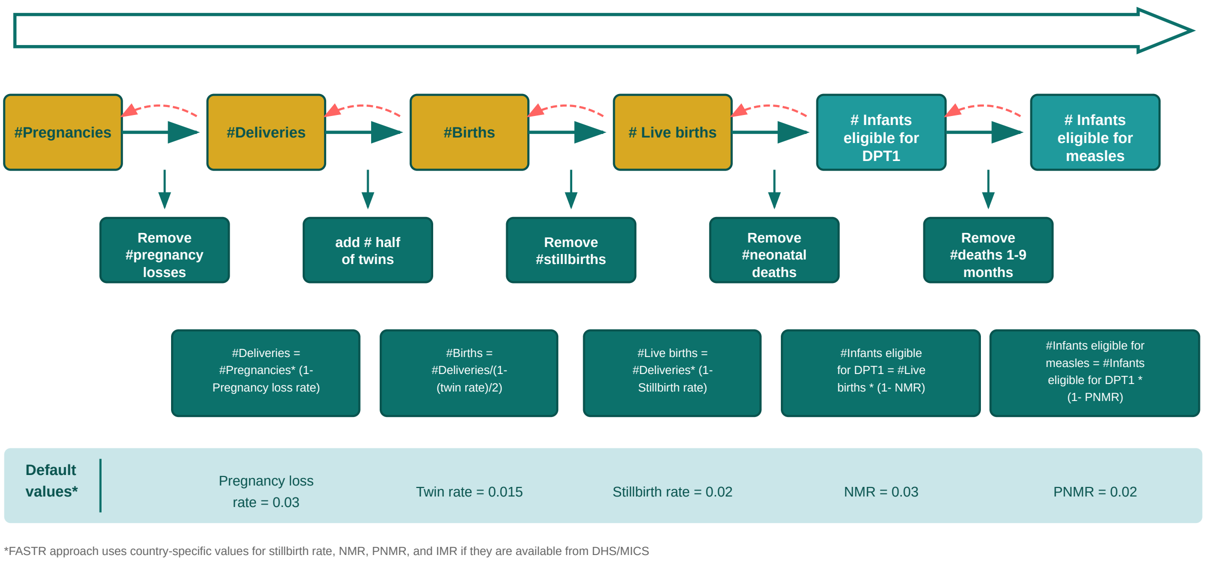
Step 2 — adjust it to the right group. 10,000 pregnant women is the right denominator for antenatal care — but **Penta1 is given to infants**, a different group. We don't need a second survey: a pregnancy *becomes* a birth *becomes* an infant, and we know roughly how many are lost at each stage. So FASTR steps the number down — minus miscarriages and stillbirths, minus newborn deaths — to about **9,100 infants**. One survey now gives the denominator for *every* indicator.

Step 3 — cross-check, and pick the best. ANC1 isn't the only place to start; deliveries, Penta1 and BCG each give their own estimate of the population, and they won't all agree. To choose, FASTR lines each estimate up against the **UN population projection** — an independent figure built without HMIS — and keeps the one that comes closest.

> ***Coverage is only as good as its denominator. That's why M5 doesn't trust one indicator — it triangulates several and lets the independent UN projection break the tie.***

M5 · THE CHAIN OF LIFE (STEP 2)

From pregnancies to infants



The **teal arrows** step *forward* — each applies one standard demographic rate (subtracting losses at that stage). The **red arrows run backward**: the same rates let you **back-calculate** the chain from any point, so FASTR can start from whichever indicator has a survey (ANC1, delivery, Penta1…) and still reach every other target population.

M6 · THE OUTPUT

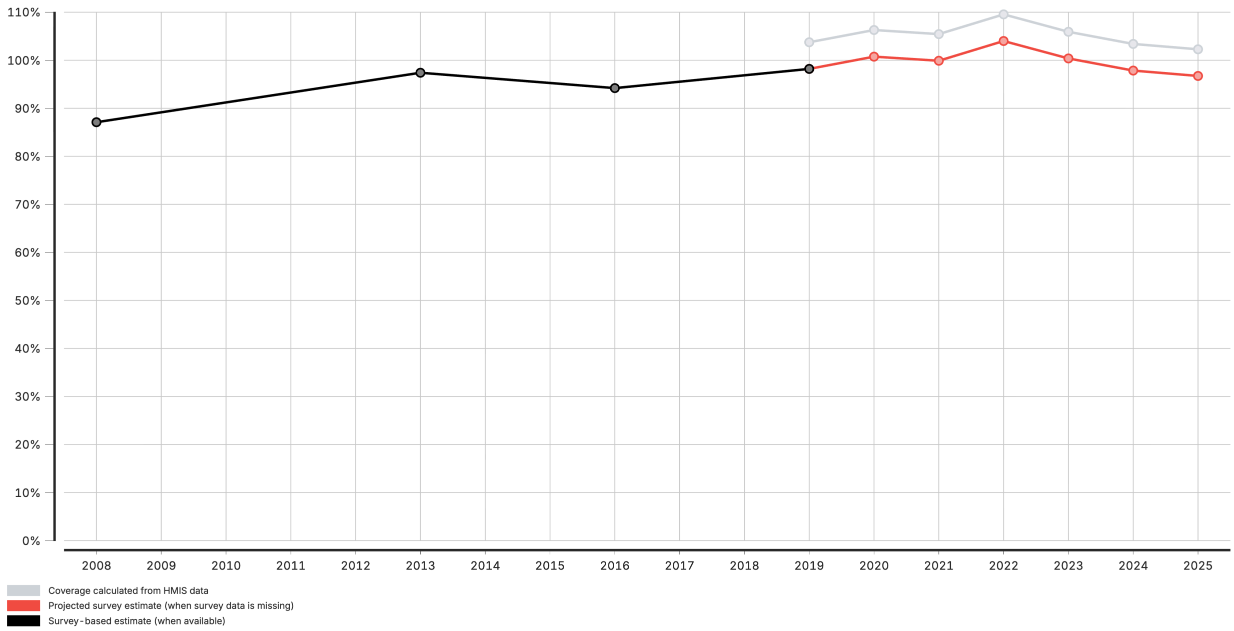
Coverage over time

Now the easy part: **coverage = HMIS count ÷ that population**, every indicator, year and area. Surveys come only every few years, so between them FASTR holds the last survey value and shifts it by however much HMIS coverage has moved — the survey sets the **level**, HMIS the **direction**.

Coverage estimates for Antenatal client 1st visit

2008 to 2025

DISCLAIMER: These results use routine data to provide rigorous, but not official estimates. They should be interpreted considering any data quality or representation limitations, including data quality findings and any other country specific factors.



Estimating service coverage from administrative data can provide more timely information on coverage trends, or highlight data quality concerns. Numerators are the volumes reported in HMIS, adjusted for data quality. Denominators are selected from UN projections, survey estimates, or derived from HMIS volume for related indicators. National projections are made by applying HMIS trends to the most recent survey data. Subnational estimates are more sensitive to poor data quality, and projections from surveys are not calculated.

MICS data courtesy of UNICEF. Multiple Indicator Cluster Surveys (various rounds) New York City, New York.

DHS data courtesy of ICF. Demographic and Health Surveys (various rounds). Rockville, Maryland.

Data for the current year reflect the period available at the time of analysis; population figures have been scaled to match the corresponding duration.

- **The lines** — **black** = HMIS coverage (count ÷ estimated population), every year; **red dots** = actual survey results; **grey** = the survey projected forward on the HMIS trend where no survey exists
- **How to read it** — where the black line and red dots sit close, the denominator is sound and the trend is trustworthy; then read the direction — rising, flat, or slipping
- **Watch for** — coverage **above 100%** is a warning sign (denominator too low or count inflated), not real over-coverage

> ***The same chart is produced at national, region and district level — read them together: a healthy national trend can still hide a struggling district.***